Cardiac Risk 3D: Interactive Coronary Artery Disease Prediction and 3D Risk Visualisation

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> **Safety notice.** This is an educational prototype for decision support. It is not a medical device and is not a substitute for formal diagnostic imaging or clinical judgement.

1. Overview

Raw risk percentages do not tell a clinician or a patient where in the heart disease may be developing. Cardiac Risk 3D predicts, from routine clinical data, (a) whether a patient has coronary artery disease (CAD) and (b) whether each of the three main coronary arteries, the left anterior descending (LAD), left circumflex (LCX) and right coronary artery (RCA), has significant stenosis. The predictions are drawn onto an interactive 3D heart: each artery is coloured by its own predicted probability, and clicking an artery shows a SHAP explanation of why the model scored it that way.

The system has three parts: a training pipeline (scikit-learn, XGBoost, SHAP), a FastAPI service that serves predictions and explanations, and a React + Three.js dashboard. Everything runs on a laptop CPU, with no dedicated GPU needed.

2. Data and preprocessing

Dataset. We use the *Extension of Z-Alizadeh Sani* dataset from the UCI Machine Learning Repository (CC BY 4.0). It contains 303 patients and 59 columns covering demographics, examination findings, ECG, laboratory values and echocardiography. The vessel labels LAD, LCX and RCA are recorded as *Stenotic* or *Normal*. The overall label `Cath` is *CAD* (216 patients) or *Normal* (87). In this file, `Cath` equals "at least one vessel stenotic" for 302 of 303 patients. Stenosis here means at least 50% narrowing of a major vessel.

Target	Source column	Positive patients
Overall CAD	`Cath`	71%
LAD stenosis	`LAD`	58%

Target	Source column	Positive patients
LCX stenosis	`LCX`	39%
RCA stenosis	`RCA`	38%

Preventing target leakage. `LAD`, `LCX`, `RCA` and `Cath` are removed from the model inputs for every target, leaving 55 input features. The code matches these names case-insensitively and an assertion stops training if any of them reappears in the feature table. This matters because the overall CAD label is almost a direct function of the three vessel labels.

Encoding and missing values. Text columns (for example `Sex`, `VHD`, `BBB`) are converted with a fixed lookup table built without looking at any label, so encoding cannot leak information into test folds. The training file has no missing cells, but users of the dashboard may leave fields blank, so a median imputer is part of every model pipeline. The imputer is fitted only on the training part of each fold. The interface marks every imputed value as "not entered".

3. Predictive modelling

Models. For each target we trained three classifiers: logistic regression (baseline), random forest and XGBoost. Only the two tree models can be served, because the explanation method (SHAP TreeExplainer) requires trees. Class weights are balanced for the forest and the logistic baseline.

Validation, nested and leak-free. With only 303 patients, a single train/test split is too noisy, and tuning on the same folds that are reported inflates results. We therefore used nested cross-validation. In each of 10 outer test folds (5-fold, repeated twice, stratified):

1. Hyper-parameters are chosen by 3-fold grid search on the outer *training* part only (random forest: `max_features` in {sqrt, 0.5} and `min_samples_leaf` in {1, 3, 5}; XGBoost: `max_depth` in {2, 3, 4} and `learning_rate` in {0.03, 0.08}).
2. The decision threshold is chosen on out-of-fold predictions of that training part, by maximising Youden's J (sensitivity + specificity - 1).
3. The untouched outer test part is scored with that model and threshold.

The reported numbers therefore describe the full procedure on unseen patients. The model family with the higher nested ROC-AUC is served; this choice is made on the same folds and is slightly optimistic, but the two families differ by less than one standard deviation. Finally, the chosen model is refit on all 303 patients, and the served threshold is re-estimated from 5-fold out-of-fold predictions on all data.

Results. Mean over 10 outer folds. Recall, precision and F1 are at the fold-tuned threshold. ROC-AUC is threshold-free and shown with its standard deviation across folds.

Target	Served model	ROC-AUC	Accuracy	Recall	Precision	F1	Served threshold
CAD	Random forest	0.924 ± 0.037	0.837	0.815	0.950	0.876	0.56
LAD	Random forest	0.855 ± 0.047	0.789	0.836	0.813	0.820	0.52
LCX	Random forest	0.730 ± 0.070	0.647	0.664	0.544	0.592	0.52
RCA	Random forest	0.729 ± 0.045	0.634	0.759	0.546	0.600	0.49

XGBoost reached similar AUCs (CAD 0.922, LAD 0.840, LCX 0.727, RCA 0.704).

What the results show.

- CAD and LAD are predicted well (AUC 0.92 and 0.86). LCX and RCA are much harder (AUC about 0.73) and show wide fold-to-fold spread.
- Threshold tuning matters most for RCA: recall rises from 0.60 at a 0.5 cut-off to 0.76, at the cost of precision (0.58 to 0.55). For the majority-positive targets CAD and LAD, tuning shifts the balance towards precision instead (CAD precision 0.90 to 0.95, recall 0.90 to 0.82). The served thresholds can be changed in `models/meta.json` if a screening setting with higher recall is preferred.
- Hyper-parameter tuning did not raise AUC beyond the untuned models. The limit appears to come from the available features, not from model capacity.
- We tested sigmoid calibration and found small gains (Brier score changes of about 0.01 or less, with CAD random forest 0.120 to 0.099). Because the gain is small and calibration would complicate the SHAP explanation, we serve the uncalibrated forests.

4. Clinical interpretability

For every prediction, the dashboard shows a per-patient SHAP explanation for the selected target (overall CAD, or LAD, LCX or RCA). The ten features with the largest absolute contribution are listed with their value and a bar: red pushes risk up, blue pushes it down. The panel states the unit of the contribution (probability for random forests, log-odds for XGBoost). Features the user did not enter are labelled "not entered, filled with typical value", and the panel warns when fewer than five fields are filled, because an explanation based on imputed values says little. SHAP values describe how the model reaches its score; they are associations, not causal effects.

5. 3D visualisation

The viewer is built with Three.js through React Three Fiber and renders in a normal browser without a GPU (pixel ratio capped at 1.5).

- **Heart model.** A stylised heart is generated in code from an implicit surface equation, with a mesh created by projecting a sphere onto the surface, and an aorta and pulmonary trunk drawn for context. We chose this over a downloaded mesh because it has no licence issues and lets us guarantee that each artery lies exactly on the surface. It is not patient-specific anatomy. A real `.glb` mesh can be loaded through a `modelUrl` option, but the artery paths would then need to be aligned by hand.
- **Artery nodes.** LAD, LCX and RCA are three separate tube meshes. Each has its own material, so every model output maps to exactly one displayed structure. Their paths are short recipes of surface points that are snapped to the heart surface automatically.
- **Colour mapping.** Probability maps from green through yellow to red, with the yellow midpoint placed at that vessel's decision threshold. Orange or red therefore means the model flags the vessel. Colours fade smoothly when inputs change.
- **Interaction.** Users rotate and zoom with the mouse or touch, hover to highlight, and click an artery to select it. Selection updates the explanation panel. Labels show each artery's name and probability.
- **Safety disclaimer.** A visible banner at the top of the interface states that predictions are for decision support and education only.

6. System architecture and integration

Component	Technology	Role
Training	scikit-learn, XGBoost, pandas	Nested CV, tuning, thresholds, saved models
Explanation	SHAP	Per-patient feature contributions
API	FastAPI	`/predict`, `/explain/{target}`, `/schema`, `/metrics`, `/examples`
Dashboard	React, React Three Fiber, Vite	Form, 3D heart, risk panel, SHAP panel

The patient form is generated automatically from the trained feature list returned by `/schema`. Each change in the form triggers a prediction after a short pause (300 ms), so the artery colours update almost immediately. The `/examples` endpoint serves three real

patients from the dataset (all arteries narrowed, LAD only, and normal) so the system can be demonstrated quickly; the interface states that these patients were part of the training data, so their predictions look better than they would for new patients.

Extensibility. Adding a clinical feature needs only a new column and a retrain, because the form and the API follow the saved feature list. A new model or target is a change to a configuration table. A new anatomical structure is one more recipe in the vessel table.

7. Reproducibility and usage

All random seeds are fixed (seed 42) and the code contains no hidden state. From the repository root: create a virtual environment, run `pip install -r requirements.txt`, place the dataset in `data/raw/`, run `python -m ml.train_tuned` (about 5 to 15 minutes on a CPU), start the API with `uvicorn backend.app:app --port 8000`, and start the dashboard with `npm install` then `npm run dev` inside `frontend/`. Trained models and `meta.json`, including all cross-validation results, are included in the repository. A faster baseline run is available with `python -m ml.train`.

8. Limitations and responsible use

- **Small data.** 303 patients, from a single source, with no external validation. The standard deviations across folds (up to 0.07 AUC for LCX) show how uncertain the estimates are.
- **Weaker vessel-level models.** LCX and RCA reach only about 0.73 AUC. A plausible explanation, which we did not test, is that the ECG and echocardiography features carry less information about these territories than about the LAD.
- **Threshold choice.** Youden's J weights sensitivity and specificity equally. A real clinical setting would set thresholds from the relative cost of missed disease and unnecessary tests.
- **Probabilities are approximate.** The forests were not recalibrated, so a displayed 70% should not be read as a 70% true chance of disease.
- **Stylised 3D model.** The heart and the artery paths are schematic. They show which vessel is flagged, not where along a vessel a narrowing sits; the dataset has no lesion locations.
- **Explanations are model-based.** SHAP reflects the model's learned associations, and correlated features can share credit. Imputed values weaken explanations.
- **Sample patients are optimistic.** They were in the training data.
- **Not for clinical use.** The system has not been clinically validated and cannot replace imaging, an angiogram or a clinician.

9. Credits and references

- Dataset: Alizadehsani, R., Roshanzamir, M. and Sani, Z. A., *Extension of Z-Alizadeh Sani dataset*, UCI Machine Learning Repository, DOI 10.24432/C5461K (CC BY 4.0).
- Software: scikit-learn, XGBoost, SHAP, pandas, FastAPI, React, Three.js, React Three Fiber and drei, Vite.
- 3D heart: generated in code (`frontend/src/components/heartMath.js`); no third-party mesh is used.